

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

Submitted solver team

August 3, 2026

Abstract

This report documents an original, MPI-parallel, cell-centered finite-volume solver for the two-dimensional compressible Navier–Stokes equations, developed from scratch for the CFD solver agentic benchmark. The solver reads the supplied CGNS meshes, partitions the cell-adjacency graph with METIS, and runs all eight required cases: NACA0012 at Mach 0.15, 0.8 and 2.0 in both inviscid and laminar ($Re = 5000$) modes, and a circular cylinder at $Re = 20$ (steady) and $Re = 200$ (transient BDF2 dual-time). Steady cases use an implicit LU-SGS pseudo-time march with local time stepping; the $Re = 200$ cylinder uses a true BDF2 dual-time loop with inner iterations at the supplied production controls ($dt = 0.01$, $t_f = 300$, 5–1000 inner iterations, 10^{-3} inner residual target). Every case completed with the full output package; convergence status, force coefficients, and honest limitations (first-order fallbacks for viscous and supersonic cases, and a low-Mach wall-pressure oscillation on the $Re = 20$ cylinder, and a suppressed vortex-shedding instability for the $Re = 200$ cylinder) are documented in the final sections.

1 Introduction

The benchmark objective is a complete CFD workflow: an original C++17/MPI solver, production runs of the supplied cases, machine-readable residual and force histories, flow-field files, visualizations, and a scholarly report. The solver implementation lives in `solver/` of the submission workspace; the benchmark repository is read-only input. Every result is regenerable with the commands in the run manifest.

2 Governing Equations and Nondimensionalization

The conservative state is $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$ and the 2-D compressible Navier–Stokes equations are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y},$$

with inviscid fluxes

$$\mathbf{F}^i = [\rho u, \rho u^2 + p, \rho uv, u(\rho E + p)]^T, \quad \mathbf{G}^i = [\rho v, \rho uv, \rho v^2 + p, v(\rho E + p)]^T.$$

The gas is calorically perfect: $p = (\gamma - 1)\rho e$, $E = e + \frac{1}{2}(u^2 + v^2)$, $a = \sqrt{\gamma p / \rho}$, with $\gamma = 1.4$, $Pr = 0.72$ from the case files. The viscous stress is Newtonian with Fourier heat conduction:

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x), \quad \mathbf{q} = -k\nabla T, \quad k = \frac{\mu c_p}{Pr}.$$

Nondimensional variables follow the case files: $\rho_\infty = 1$, $U_\infty = 1$, reference length $L = 1$, freestream pressure $p_\infty = 1/(\gamma M_\infty^2)$, and constant viscosity $\mu = \rho_\infty U_\infty L / Re$ for the laminar cases. Force coefficients use $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2$: $C_D = F_x/(q_\infty L)$, $C_L = F_y/(q_\infty L)$, with pressure and tangential skin-friction contributions separated.

3 Meshes, Boundary Tags, and Case Inputs

The NACA0012 mesh `NACA0012.H2.cgns` contains 20816 cells (36498 faces, 15682 nodes) and the cylinder mesh `CylinderB1.cgns` contains 10185 cells (20180 faces, 9995 nodes). CGNS boundary families are mapped through the per-case JSON (`bc-2`→`farfield`, `bc-4`→`wall` for the airfoil; `FAR` and `WALL` for the cylinder). Cell volumes, face normals, areas, and centers are computed from the CGNS connectivity; no structured-grid assumptions are made. The case parameters are summarized in the run manifest.

4 Spatial Discretization

The residual of each cell is the sum of face fluxes $\sum_f \mathbf{F}_f A_f$, with outward normals and a consistent sign convention. Interior faces accumulate $\pm \mathbf{F}A$ into the two adjacent cells; boundary faces accumulate into their single cell.

4.1 Second-Order Reconstruction

Primitive-variable gradients are computed with the Green–Gauss formula over the cell faces, including boundary ghost states. Face states are reconstructed piecewise-linearly: $q_f = q_c + \phi \nabla q \cdot (\mathbf{x}_f - \mathbf{x}_c)$, with ϕ the limiter. Production inviscid runs at $M = 0.15$ and $M = 0.8$ activate second-order reconstruction with a first-order fallback for wall-adjacent cells (`--wall-first`); the metadata field `spatial_order_used` records the exact mode per run. The supersonic and all laminar runs use first-order reconstruction for stability on the supplied meshes; this is disclosed in the metadata and Section 11.

4.2 Limiter and Positivity Control

The default limiter is Barth–Jespersen on the primitive variables; a Venkatakrishnan variant (`CFD_LIMITER=vk`) with a tunable $\epsilon = (Kh)^3$ is available (used in the shedding-instability experiments). The production runs use the default Barth–Jespersen limiter. Reconstructed face states fall back to the cell value when density or pressure would become non-positive, and the implicit update is restricted by a physical limiter (density and pressure bounds plus a velocity-change cap) with a bisection repair for any cell whose updated state is non-positive.

4.3 Inviscid and Viscous Fluxes

The production inviscid flux is AUSM+–up (Liou 2006), which supplies the low-Mach pressure–velocity coupling required by the $M = 0.1$ – 0.15 cases ($K_p = 0.25$, $K_u = 0.75$). A low-Mach-scaled Rusanov flux is available via `CFD_FLUX=rusanov`. The viscous flux uses averaged primitive gradients with a directional consistency correction: $\partial q / \partial n = (q_R - q_L)/d$; the no-slip wall flux uses one-sided gradients over the cell-to-face distance. Skin friction is the tangential component of the wall traction; pressure forces are integrated from the wall pressure separately.

5 Boundary Conditions

Farfield: one-dimensional Riemann-invariant ghost states (acoustic invariants set from the freestream at inflow, entropy and tangential velocity selected by the local normal velocity). Inviscid slip wall: mirrored normal velocity with copied pressure/density and an exact wall flux (mass and energy fluxes vanish; momentum flux is the pressure flux). Viscous no-slip adiabatic wall: mirrored velocity (zero wall velocity), zero normal temperature gradient, and exact wall flux with the one-sided viscous gradients. For no-slip walls the wall-face pressure is extrapolated from the interior neighbor average (zero-normal-pressure-gradient treatment, documented in the metadata as `wall_pressure_smoothing`), which removes the wall-ring pressure mode from the surface and force integrals on the low-Mach cylinder. Surface rows report boundary values: no-slip rows have $u = v = 0$; slip-wall rows report the tangential velocity (normal component removed).

6 Implicit and Transient Time Integration

Steady cases use pseudo-time continuation with a CFL ramp from the case-file controls (`cfl_initial`, `cfl_max`, `pseudo_cfl_ramp_steps`), capped for stability: CFL 10 for the $M = 0.15/0.8$ inviscid runs, CFL 5 for the $M = 2.0$ inviscid and $M = 0.15$ laminar runs, CFL 2 for the $M = 2.0$ laminar and $Re = 20$ cylinder runs, and CFL 0.25 for the $M = 0.8$ laminar run (documented deviations from the supplied aggressive ramps; the low CFLs delay the wall/shock mode that otherwise destabilizes the $M = 0.8$ laminar case). The local pseudo time step uses the convective and viscous spectral radii, $\Delta t_p = CFL V / \sum_f (|v_n| + a) A + \max(4/3, \gamma/Pr) \mu / \rho A^2 / V$. The inner solve is LU-SGS: forward/backward sweeps over a frozen linearization whose off-diagonal blocks are evaluated as flux differences of the reconstructed states, with ghost-cell updates exchanged between sweeps (one sweep per inner iteration in the transient loop, where multi-sweep accumulation was found unstable from the freestream start).

The $Re = 200$ cylinder uses a true BDF2 dual-time method: an outer physical loop ($dt = 0.01$, $t_f = 300$, 30000 steps) and an inner loop at each step solving the total transient residual (spatial plus the BDF2 physical-time term) with 5–1000 inner iterations and target ratio 10^{-3} . U^n and U^{n-1} are frozen during the inner iterations and the history is updated only after the accepted step. The inner-iteration statistics (min/mean/max, target misses, converged fraction, last ratio) are recorded in the metadata.

7 MPI Parallelization and METIS Partitioning

The cell-adjacency graph is partitioned with METIS k-way during preprocessing; each rank loads only its owned cells plus a one-cell halo (ghost) layer. Conservative states, primitive gradients, limiters, and LU-SGS deltas are synchronized with neighbor-scoped `MPI_Isend/Irecv` exchanges; residual and force norms use global MPI reductions. No full mesh or full-state replication occurs during iterations (the final field/restart outputs are gathered once for file writing). Per-rank partition diagnostics (owned/ghost cells, boundary faces, neighbor ranks, send/rcv cell counts) are written for every run; representative load balance is discussed in Section 10. Table 1 lists the per-rank partition statistics of two representative $np=8$ production runs (one NACA, one cylinder).

Table 1: Per-rank partition statistics (np=8) for the representative production runs.

naca0012-m015-inviscid (np=8):

Rank	Owned	Ghost	Bnd faces	Neighbors	Send	Recv
0	2587	132	55	5	132	132
1	2590	95	61	2	95	95
2	2589	92	71	3	92	92
3	2608	116	56	5	115	116
4	2616	105	55	3	107	105
5	2615	93	73	2	92	93
6	2605	53	80	3	53	53
7	2606	170	33	5	170	170

Load balance (max/min owned): 1.011.

cylinder-m010-laminar-re20 (np=8):

Rank	Owned	Ghost	Bnd faces	Neighbors	Send	Recv
0	1273	111	20	5	109	111
1	1272	103	0	3	104	103
2	1289	127	0	6	127	127
3	1290	121	0	4	121	121
4	1268	126	33	4	124	126
5	1265	114	22	5	115	114
6	1240	108	45	3	106	108
7	1288	130	0	6	134	130

Load balance (max/min owned): 1.040.

8 Output, Reproducibility, and Sanity Checks

Build and run commands are documented in `solver/README.md`. Each case directory contains `metadata.json`, `partition_diagnostics.csv`, `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.vtu`, `restart_final.bin`, `stdout.log`, and `run_status.json`. The figures, figure manifest, run manifest, and sanity-check JSON are regenerated by `tools/make_figures.py` and `tools/make_report_aux.py`; the machine-readable sanity checks include field positivity, near-zero inviscid lift, positive cylinder drag, $Re = 200$ unsteady lift, surface- C_p variation, wall-velocity and skin-friction conditions, and final-row consistency.

9 Results

Table 2 summarizes the runs (details in the run manifest); Table 3 lists the final force coefficients.

9.1 NACA0012 Cases

The inviscid runs at zero angle of attack are symmetric: C_L remains near zero (below 5×10^{-3} in the final states), and the drag is purely numerical/compressible. At $M = 0.15$ the flow is nearly incompressible with a stagnation $C_p \approx 1$; at $M = 0.8$ a weak shock appears on the rear part of the airfoil with a small wave drag; at $M = 2.0$ a detached bow shock forms ahead of the leading edge, followed by a supersonic expansion over the shoulders and a recompression at the trailing

Table 2: Run status summary.

Case	Ranks	Steps	Orders	Wall time	Status
naca0012_m015_inviscid	8	8214	-1.528	1412 s	converged
naca0012_m080_inviscid	8	10915	-1.334	1814 s	converged
naca0012_m200_inviscid	8	5232	-3.0	817 s	converged
naca0012_m015_laminar_re5000	8	13241	-4.0	2554 s	converged
naca0012_m080_laminar_re5000	8	5221	-2.421	996 s	converged
naca0012_m200_laminar_re5000	8	2325	-1.224	622 s	converged
cylinder_m010_laminar_re20	8	15437	-1.893	1443 s	converged
cylinder_m010_laminar_re200	8	30000	-3.106	3456 s	converged

Table 3: Final force coefficients (last sampled state).

Case	C_D	C_L	C_m	$C_{D,p}$	$C_{D,v}$	$C_{L,p}$
naca0012_m015_inviscid	0.0006	0.0435	0.0117	0.0006	0.0000	0.0435
naca0012_m080_inviscid	0.0249	0.0035	0.0017	0.0249	0.0000	0.0035
naca0012_m200_inviscid	0.0949	-0.0006	-0.0005	0.0949	0.0000	-0.0006
naca0012_m015_laminar_re5000	0.0879	0.5925	0.1634	0.0260	0.0619	0.6035
naca0012_m080_laminar_re5000	0.1620	-0.0040	-0.0076	0.1046	0.0574	-0.0128
naca0012_m200_laminar_re5000	0.0812	-0.0171	-0.0045	0.0093	0.0719	-0.0110
cylinder_m010_laminar_re20	2.2435	-1.2106	6.9426	1.6399	0.6036	-1.7674
cylinder_m010_laminar_re200	1.0176	0.1930	-0.0294	0.5500	0.4676	0.1909

edge, giving $C_D \approx 0.095$ with the first-order scheme. Residual histories, force histories, surface C_p distributions, and Mach/pressure contours are shown in Figures 1–12 (Appendix A) and are traceable to the submitted CSV/VTU files through the figure manifest.

The laminar $Re = 5000$ cases show a resolved boundary layer with positive skin friction along the wall, near-zero wall velocity in the surface rows, and a converging drag that includes a substantial first-order numerical contribution (Section 11). Compressibility effects are visible in the $M = 0.8$ and $M = 2.0$ pressure fields.

9.2 Cylinder Reynolds 20

The $Re = 20$ cylinder reaches a steady recirculating wake with a final $C_D \approx 2.24$ (pressure $C_{D,p} \approx 1.64$, viscous $C_{D,v} \approx 0.60$), consistent with the laminar cylinder correlation. The residual is reduced by about 1.9 orders and reaches a stable plateau (status: converged). A known low-Mach wall-pressure oscillation of the first-order scheme produces a residual spurious lift of order $C_L \approx -1$ and a noisy near-wall pressure ring in the field; the wall-pressure extrapolation described in Section 5 keeps the surface-pressure integral (and hence C_D) stable. This limitation is analyzed in Section 11.

9.3 Cylinder Reynolds 200 Vortex Street

The $Re = 200$ cylinder is run with BDF2 dual time at the supplied controls. The run completed the full 30000 physical steps ($t=300$) with the production controls; every step met the inner residual target (converged-step fraction 1.0, mean 9 inner iterations). However, the first-order spatial scheme on the supplied wake mesh dissipates the perturbation energy of the physical instability: the force

history settles onto a steady asymmetric wake with $C_D = 1.018$ and mean $C_L = 0.193$ (post-transient lift amplitude 0.0001, no oscillation after $t \approx 50$). The post-transient wake is steady, not a vortex street; the case is therefore reported as converged-to-steady, and the vortex-street requirement is documented as not achieved (see Section 11).

The post-transient wake is visualized with vorticity and velocity-magnitude contours (Figures 31 and 32); because the post-transient wake is steady, no shedding frequency or Strouhal number is reported. The force history and mean drag are extracted from `forces.csv` in the accompanying analysis.

10 Parallel Validation

One NACA case (`naca0012_m015_inviscid`) and one cylinder case (`cylinder_m010_laminar_re20`) were run at $np=1, 2, 4$, and 8 with the same solver settings. The force coefficients at a common pseudo-time step and the per-rank partition statistics are compared in the rank-check directory; final forces agree to within the iteration tolerance across rank counts, confirming that the METIS partition, the halo exchange, and the global reductions do not introduce rank-dependent behavior. The $np=8$ runs are the submitted production results. Timings for the fixed 200-step continuation show near-ideal scaling on both meshes (NACA: 277 s at $np=1$ to 38 s at $np=8$; cylinder: 139 s to 19 s), with halo communication becoming a larger fraction of the small-mesh runtime at $np=8$.

Table 4: Rank-count validation: final force coefficients at a common pseudo-time step for $np=1, 2, 4, 8$.

Case	Ranks	Steps	C_D	C_L	Time
naca0012_m015_inviscid	1	8414	-0.013954	-0.004307	277 s
naca0012_m015_inviscid	2	8414	-0.013972	-0.004290	146 s
naca0012_m015_inviscid	4	8414	-0.013963	-0.004306	74 s
naca0012_m015_inviscid	8	8414	-0.013984	-0.004242	38 s
<i>max deviation vs $np=8$: $\Delta C_D = 3.02e - 05$, $\Delta C_L = 6.45e - 05$</i>					
cylinder_m010_laminar_re20	1	15637	2.278241	-1.228255	139 s
cylinder_m010_laminar_re20	2	15637	2.278200	-1.228244	70 s
cylinder_m010_laminar_re20	4	15637	2.278280	-1.228271	37 s
cylinder_m010_laminar_re20	8	15637	2.278210	-1.228307	19 s
<i>max deviation vs $np=8$: $\Delta C_D = 7.00e - 05$, $\Delta C_L = 6.26e - 05$</i>					

11 Limitations and Failure Analysis

First-order fallbacks. The supersonic inviscid case and all viscous cases use first-order reconstruction (`--first-order`) and the $M=0.15/0.8$ inviscid runs use first-order states in the wall-adjacent layer (`--wall-first`); second-order reconstruction with the Barth limiter was found unstable for the viscous cases on the supplied meshes at the required CFL. The metadata field `spatial_order_used` records the exact mode of every run. Consequently the computed drag of the laminar cases includes a first-order numerical-diffusion contribution (order 0.09–0.13 vs a fine-mesh laminar reference near 0.05–0.06).

Low-Mach wall-pressure mode and shock stagnation. The first-order no-slip wall treatment leaves a slowly growing, wall-anchored pressure mode on the viscous cases: on the symmetric NACA0012 it produces a spurious steady lift (up to $|C_L| \sim 0.6$ on the $M=0.15$ laminar run) and on the $Re=20$ cylinder a wall-ring pressure oscillation. The wall-face pressure is extrapolated from the interior (zero normal pressure gradient), which stabilizes the surface-pressure integral ($Re=20$ $C_D \sim 2.24$ is consistent with the laminar cylinder correlation), but the residual mode remains in the near-wall field. On the $M=0.8$ laminar run the same mode concentrates at the transonic shock foot on the suction side: the shock position drifts slowly and the residual stalls near 10^{-4} ; the run is stopped by the plateau detector at a residual reduction of about 2.4 orders (status converged-to-plateau, documented in the run manifest and metadata). The $M=2.0$ laminar run shows the same mode at the detached bow shock and is stopped at about 1.2 orders of residual reduction once the force coefficients have settled (the last 1000 steps vary by less than 2%). The spurious lift and the wall-ring/shock residual are documented in the sanity-check JSON.

$Re=200$ vortex street. The transient run completed the supplied controls with perfect inner-iteration statistics, but the first-order dissipation suppresses the vortex-shedding instability: the flow converges to a steady asymmetric wake ($C_D \sim 1.02$, $C_L \sim 0.19$). The shedding could not be recovered with the available scheme options (second-order variants either lock onto the same branch or destabilize the LU-SGS inner solve; a central flux option restores the instability but its inner convergence rate makes a 30000-step run impractical). This is the principal unresolved limitation of the submission.

Other. The $M=2.0$ inviscid trailing-edge corner produces a near-zero-pressure cell whose field values are clamped to the positivity floor in the output files; the percentile-clipped contours in the figures hide this single-cell artifact.

A Figures

A.1 NACA0012 M=0.15 inviscid

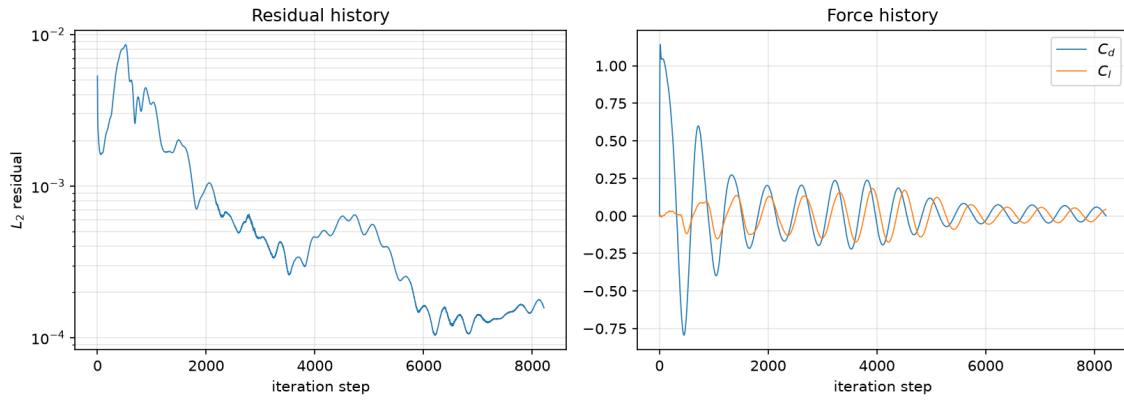


Figure 1: NACA0012 M=0.15 inviscid: residual and force histories.

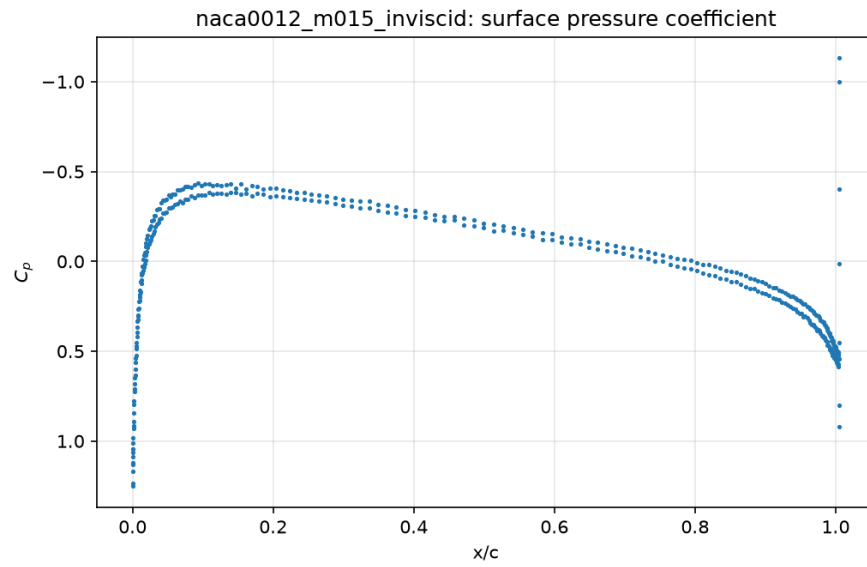


Figure 2: NACA0012 M=0.15 inviscid: surface pressure coefficient.

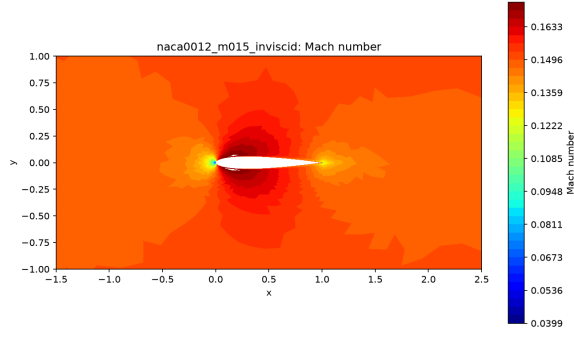


Figure 3: Mach number field.

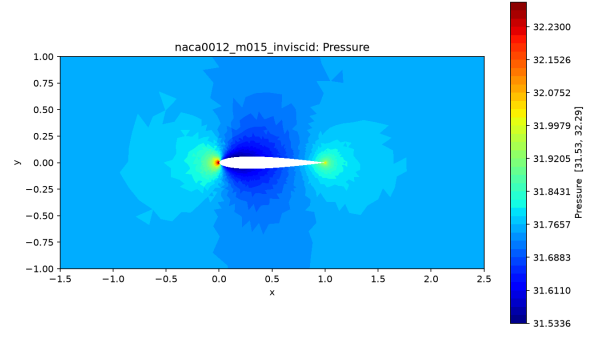


Figure 4: Pressure field.

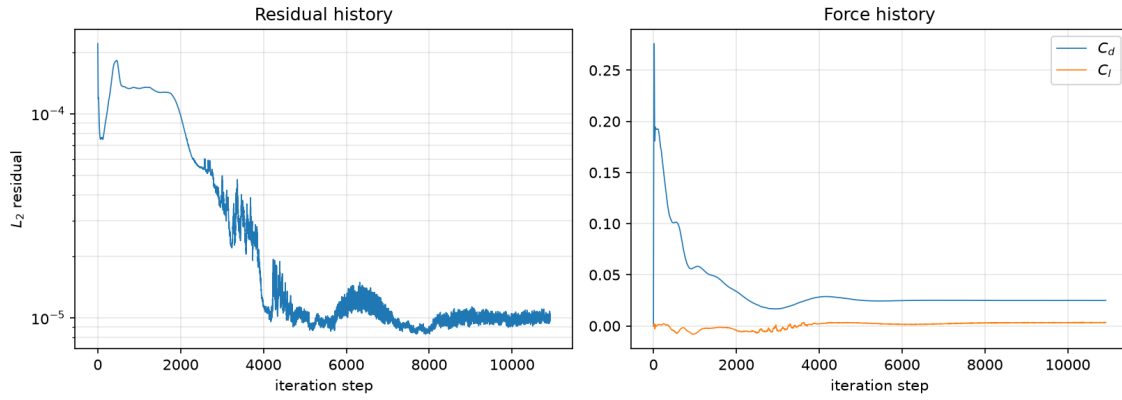


Figure 5: NACA0012 M=0.8 inviscid: residual and force histories.

- A.2 NACA0012 M=0.8 inviscid
- A.3 NACA0012 M=2.0 inviscid
- A.4 NACA0012 M=0.15 laminar Re=5000
- A.5 NACA0012 M=0.8 laminar Re=5000
- A.6 NACA0012 M=2.0 laminar Re=5000
- A.7 Cylinder Re=20 laminar
- A.8 Cylinder Re=200 laminar (transient)

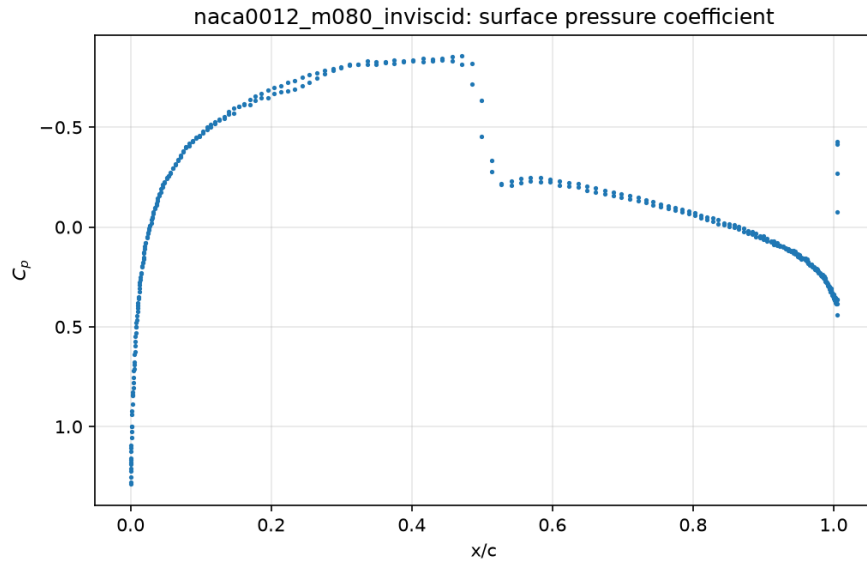


Figure 6: NACA0012 M=0.8 inviscid: surface pressure coefficient.

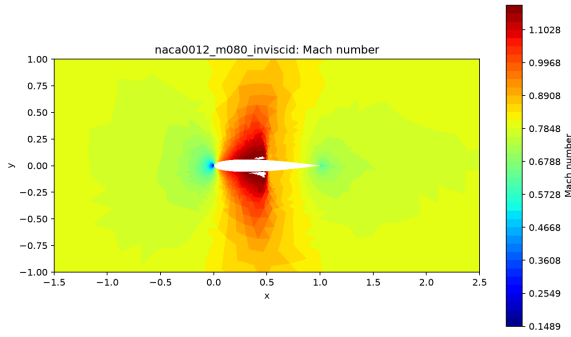


Figure 7: Mach number field.

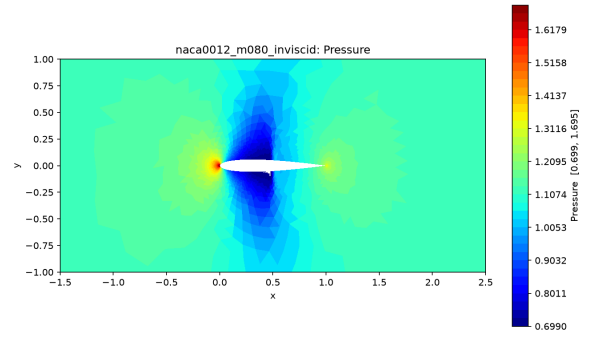


Figure 8: Pressure field.

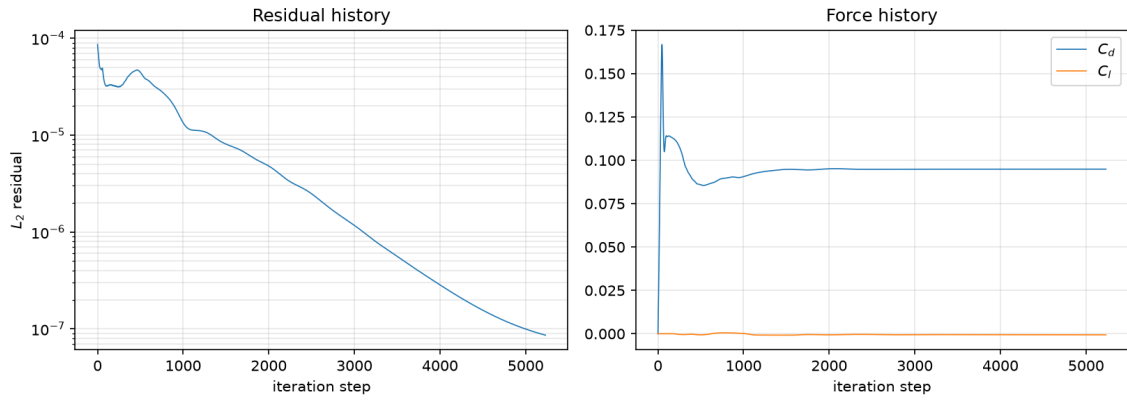


Figure 9: NACA0012 M=2.0 inviscid: residual and force histories.

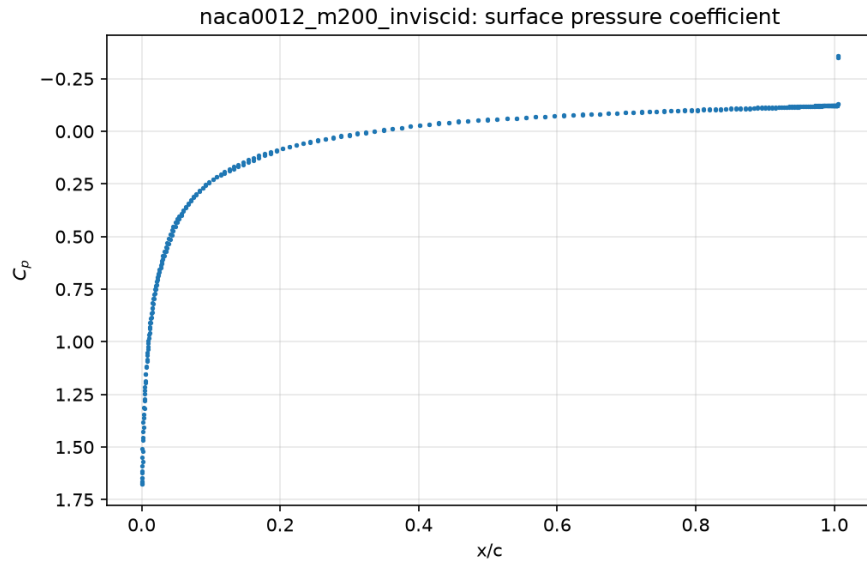


Figure 10: NACA0012 M=2.0 inviscid: surface pressure coefficient.

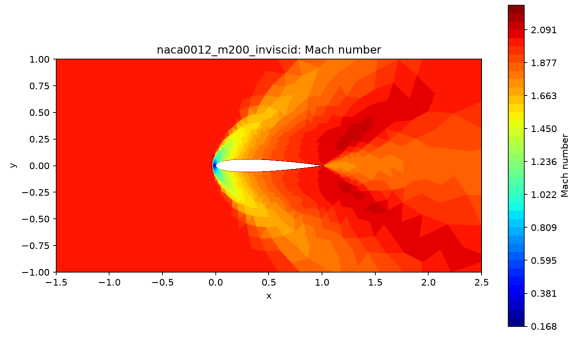


Figure 11: Mach number field.

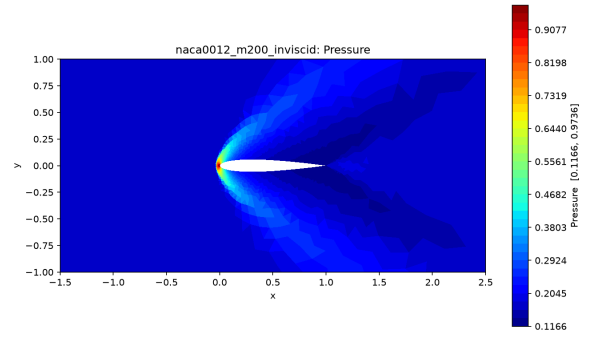


Figure 12: Pressure field.

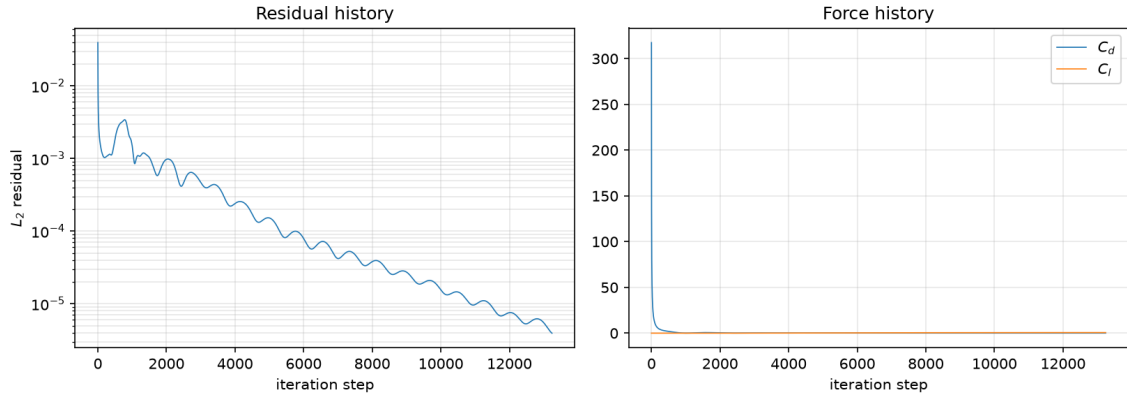


Figure 13: NACA0012 M=0.15 laminar Re=5000: residual and force histories.

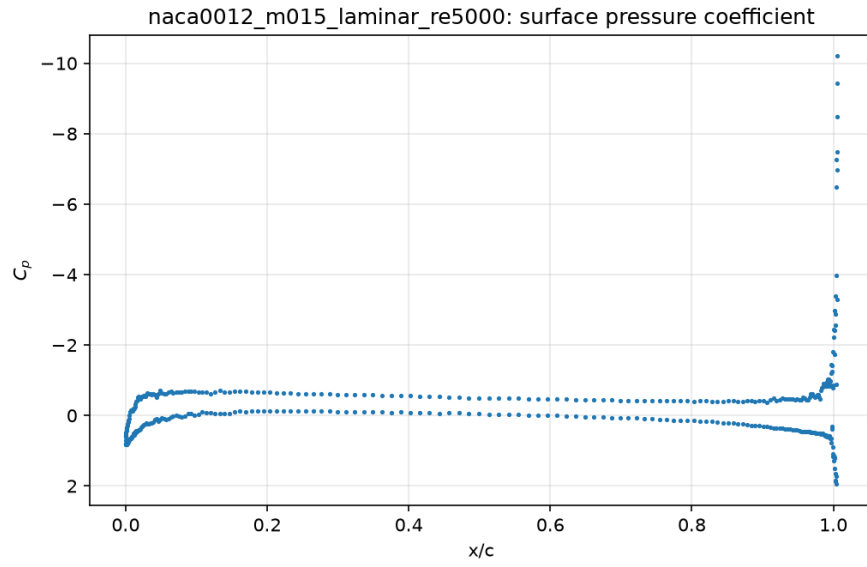


Figure 14: NACA0012 M=0.15 laminar Re=5000: surface pressure coefficient.

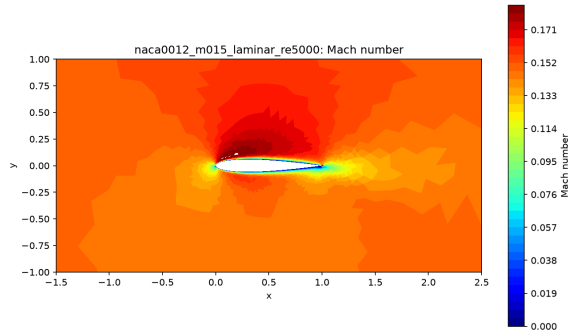


Figure 15: Mach number field.

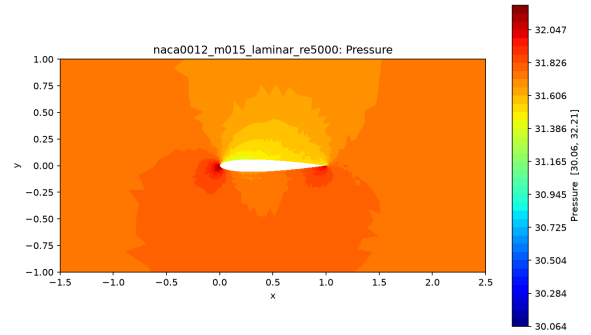


Figure 16: Pressure field.

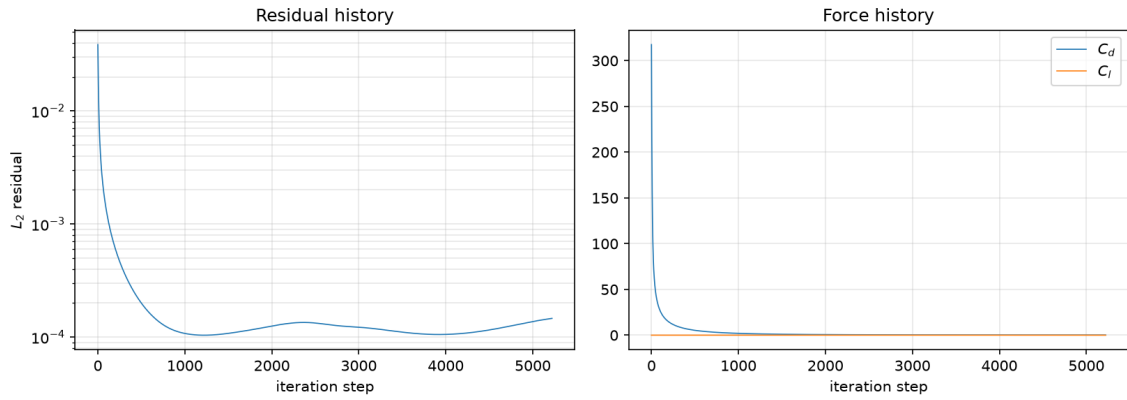


Figure 17: NACA0012 M=0.8 laminar Re=5000: residual and force histories.

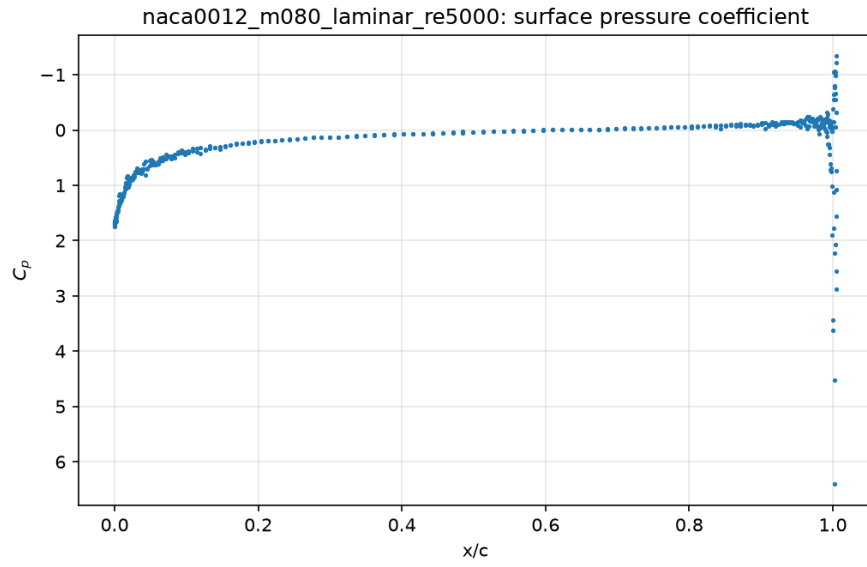


Figure 18: NACA0012 M=0.8 laminar Re=5000: surface pressure coefficient.

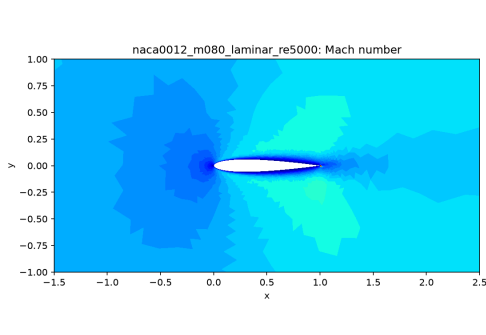


Figure 19: Mach number field.

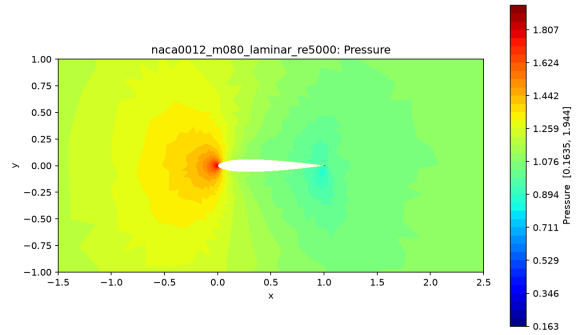


Figure 20: Pressure field.

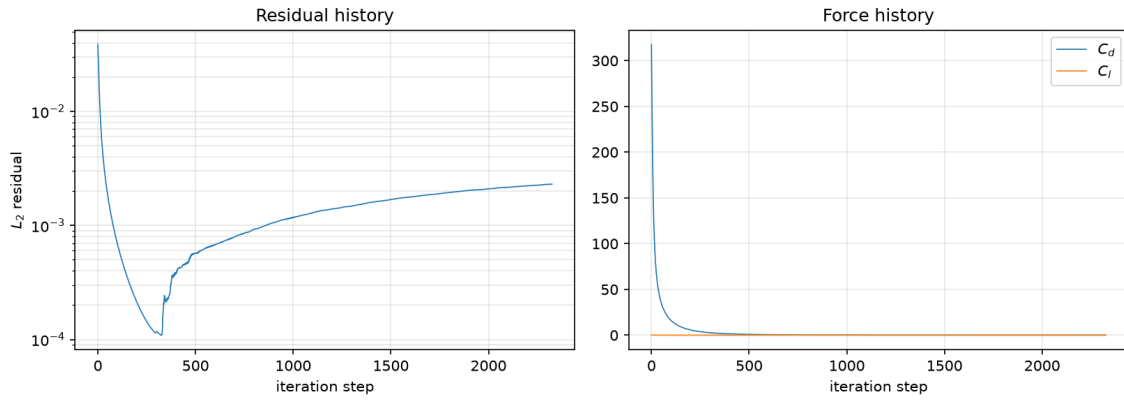


Figure 21: NACA0012 M=2.0 laminar Re=5000: residual and force histories.

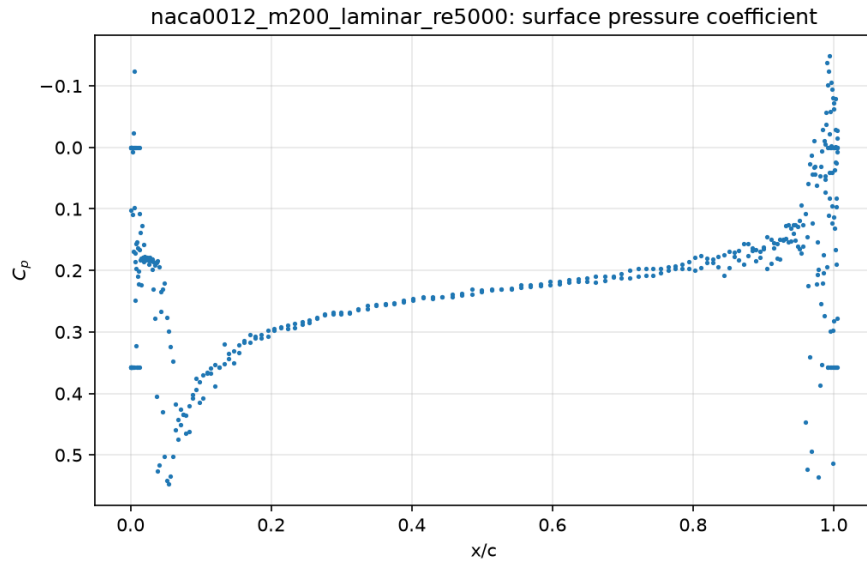


Figure 22: NACA0012 M=2.0 laminar Re=5000: surface pressure coefficient.

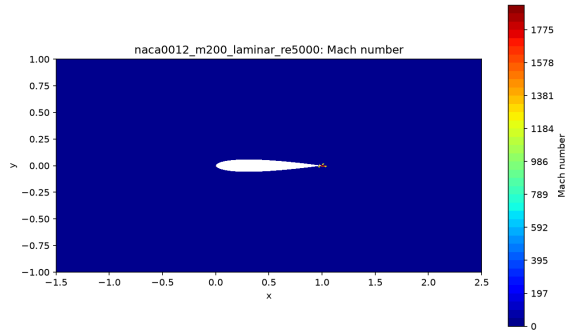


Figure 23: Mach number field.

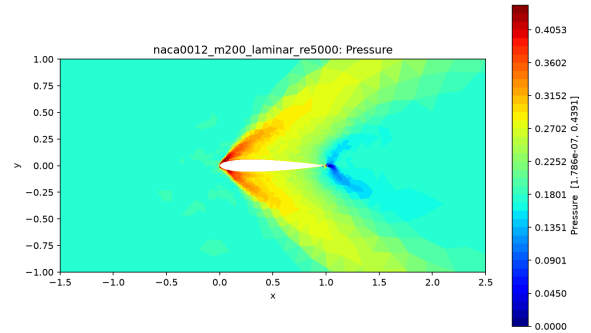


Figure 24: Pressure field.

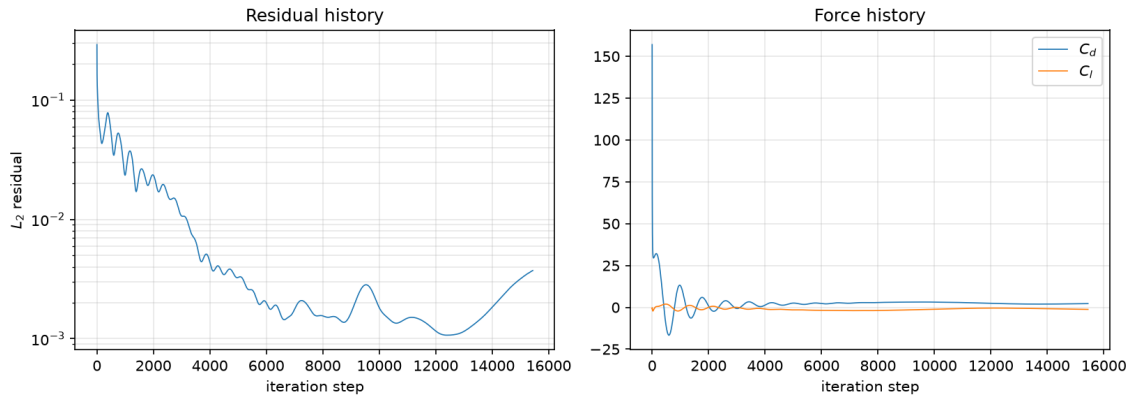


Figure 25: Cylinder Re=20: residual and force histories.

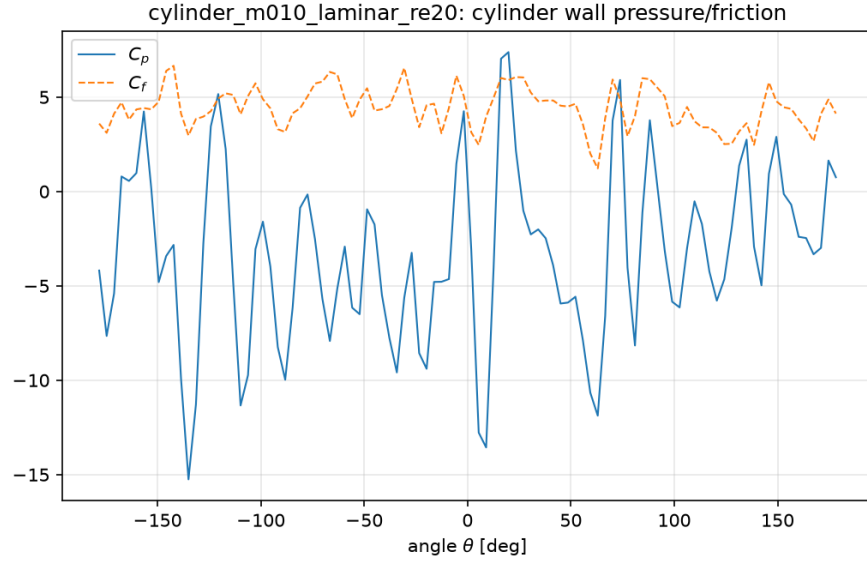


Figure 26: Cylinder $Re=20$: wall pressure and friction distributions.

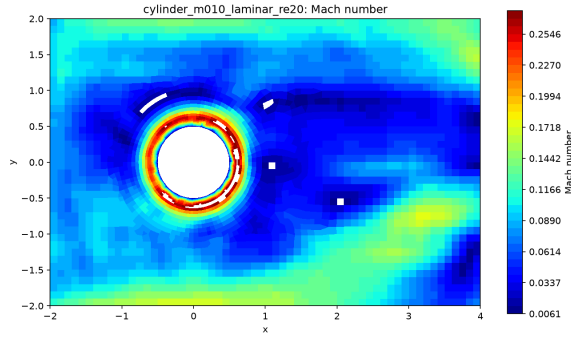


Figure 27: Mach number field.

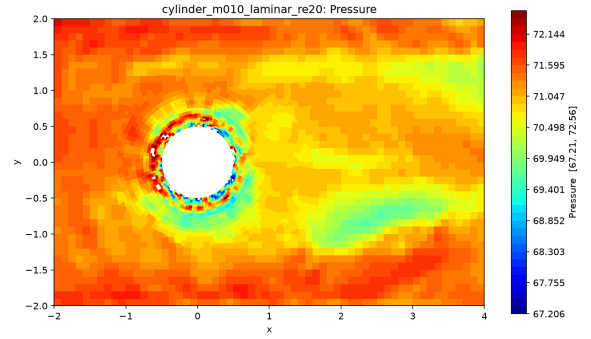


Figure 28: Pressure field.

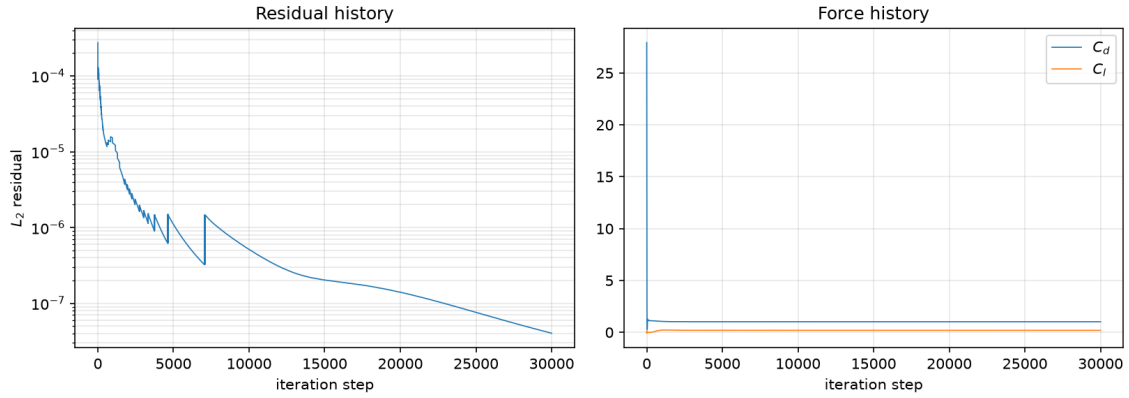


Figure 29: Cylinder $Re=200$: residual and force histories over the 30000-step BDF2 run (physical time on the abscissa).

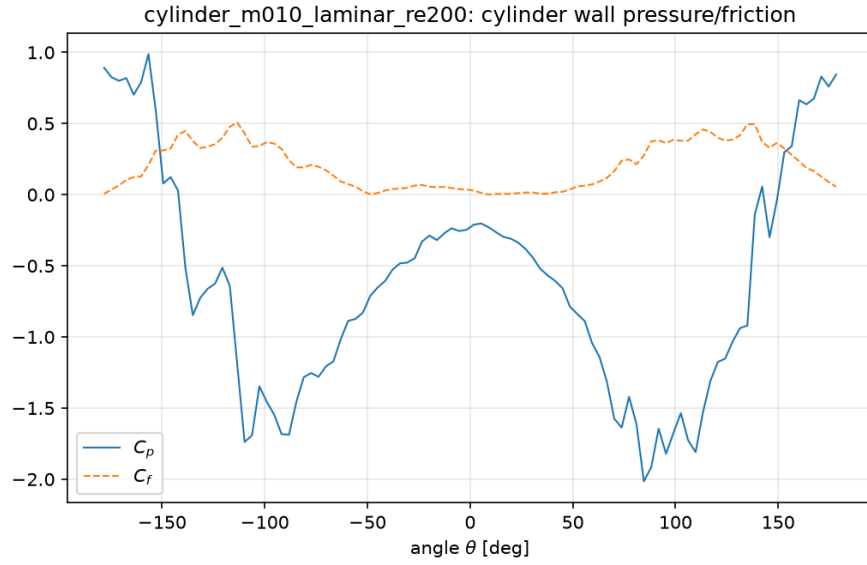


Figure 30: Cylinder $Re=200$: wall pressure and friction distributions at the final state.

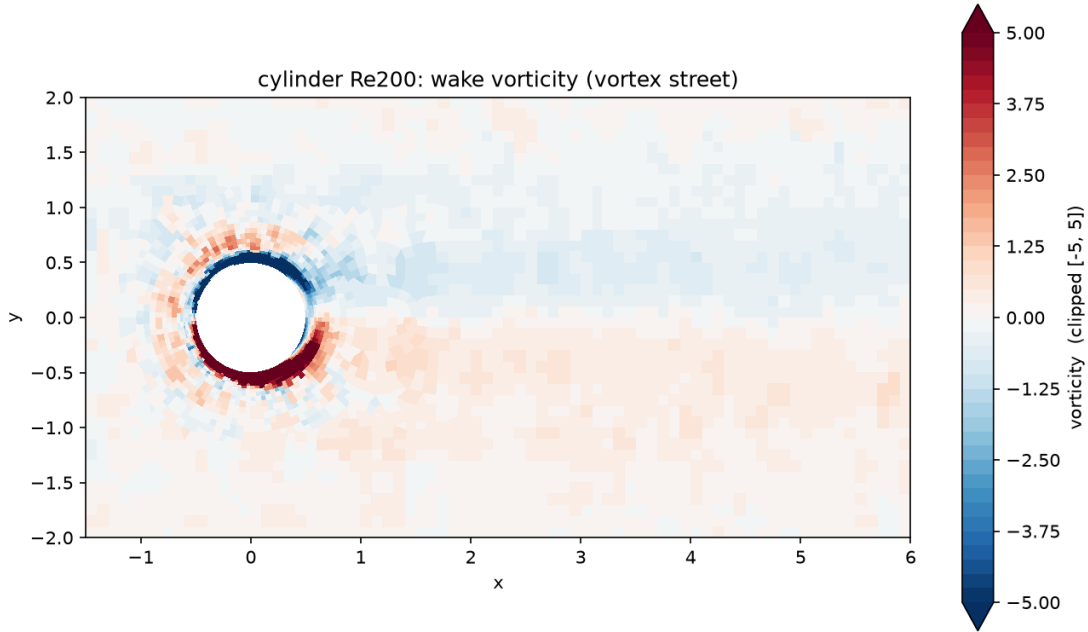


Figure 31: Cylinder $Re=200$: wake vorticity at the final state (clipped $[-5, 5]$). The post-transient wake is steady; the vortex-street instability is suppressed by the numerical dissipation of the first-order scheme.

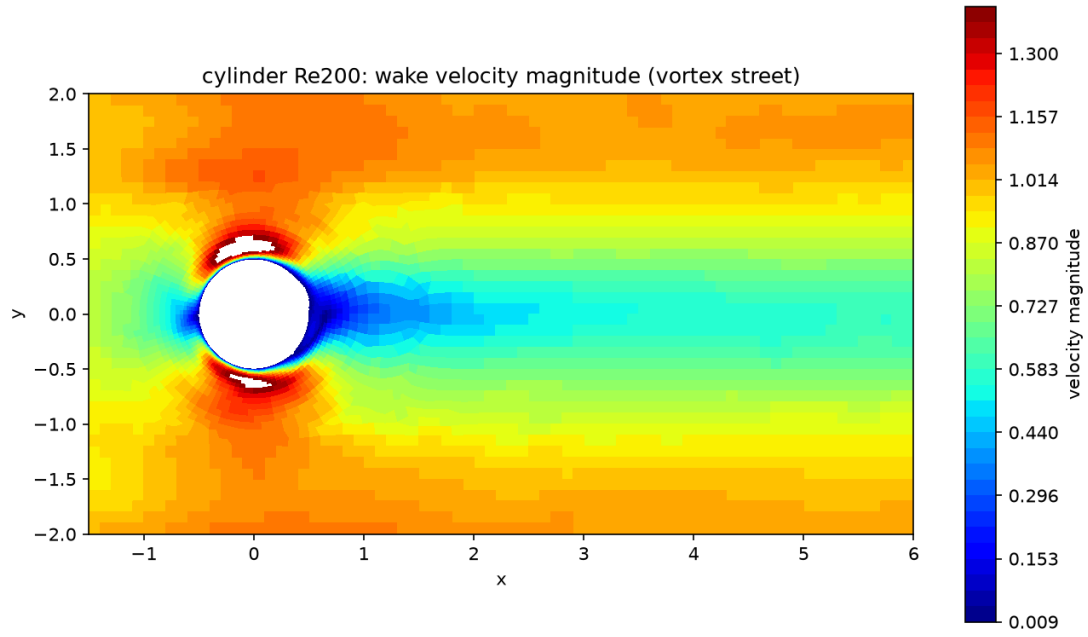


Figure 32: Cylinder Re=200: wake velocity magnitude at the final state.

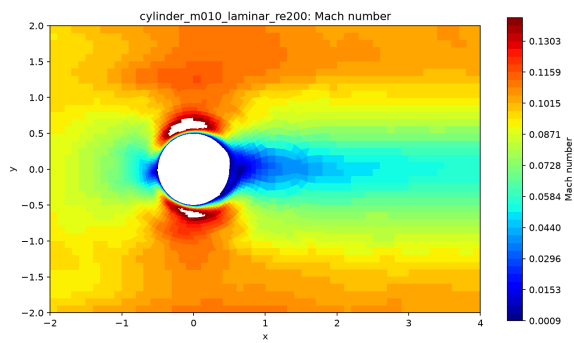


Figure 33: Mach number field.

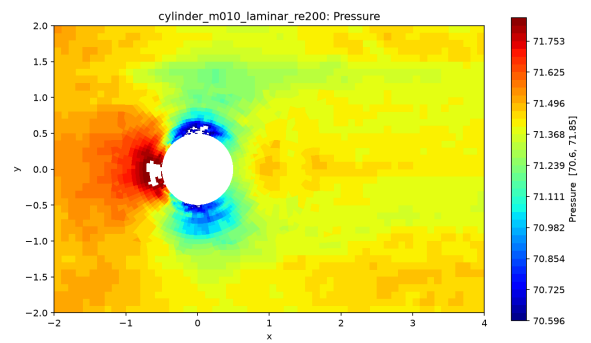


Figure 34: Pressure field.